MoCA-Dialog: A Benchmark for Fine-Grained Evaluation of Large Language Models in Clinical Cognitive Assessment Dialogues

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***Abstract*—Current benchmarks for evaluating large language models (LLMs) in medicine primarily focus on static question-answering, neglecting the dynamic, interactive nature of many clinical workflows. This is particularly evident in cognitive assessments, where nuanced dialogue and multimodal interpretation are critical, yet no specialized benchmark exists to evaluate these capabilities. To address this gap, we introduce MoCA-Dialog, the first large-scale, multimodal benchmark featuring 5000 high-fidelity simulated dialogues for the Montreal Cognitive Assessment (MoCA). The benchmark includes tasks of increasing complexity: accurate scoring, cognitive profile generation, and clinical error attribution, allowing for a fine-grained evaluation across seven cognitive domains. Our comprehensive evaluation of state-of-the-art multimodal LLMs reveals significant performance disparities; while models excel at simple recall tasks, they consistently fail in domains requiring executive function and abstract reasoning. A novel, clinically-driven error analysis further indicates that these failures stem not from knowledge deficits, but from fundamental difficulties in interpreting nuanced cues and applying domain-specific reasoning. MoCA-Dialog provides a crucial tool for assessing the clinical readiness of LLMs and highlights that future progress depends on enhancing their core reasoning and interpretive abilities, not just expanding their knowledge base.**

***Index Terms*—cognitive assessment, MoCA, large language model, medical benchmark**

I. Introduction

Large language models (LLMs) such as the GPT series [1], Gemini [2], and Llama [3] have been successfully applied to a variety of biomedical tasks, including, but not limited to, question answering [4], [5], clinical trial matching [6], and medical document summarization [7]. However, most of these tasks have a limited evaluation of the models' ability to engage in dynamic, interactive reasoning, as demonstrated by the commonly used medical benchmarks such as MedQA [4], PubMedQA [8], and MedMCQA [5]. While these benchmarks are vital for assessing domain knowledge and descriptive reasoning from static text, they do not capture the procedural and interactive nature of many real-world clinical workflows.

In clinical practice, particularly in neurology and psychiatry, cognitive assessments are a cornerstone of diagnostic evaluation. The Montreal Cognitive Assessment (MoCA) [9] is a widely used screening tool that requires a clinician to guide a patient through a series of complex, multimodal tasks. However, no existing benchmark is designed to evaluate the capabilities of LLMs in this critical, interactive, and procedurally complex clinical setting. Successfully administering and interpreting the MoCA involves more than simple question-answering; it demands the ability to deliver precise instructions, understand nuanced verbal responses—including hesitations, self-corrections, and non-literal language—and integrate information from different modalities.

To this end, we propose **MoCA-Dialog**, a first-of-its-kind, multimodal benchmark focused on evaluating the fine-grained capabilities of LLMs in conducting the MoCA assessment through simulated clinical dialogue. MoCA-Dialog is a multimodal, dialogue-based framework. Each instance in MoCA-Dialog consists of a complete patient-clinician dialogue script, a corresponding answered image for visual tasks (e.g., the clock drawing test) and the audio of the patient's responses. Our evaluation results show that even state-of-the-art multimodal LLMs exhibit significant weaknesses, particularly in tasks requiring abstract reasoning and executive function, revealing critical gaps in their clinical reasoning abilities.

In summary, the contributions of our study are threefold:

- We manually curated and release MoCA-Dialog, a novel, high-fidelity, and multimodal dataset of over 5000 clinical dialogues for evaluating the capabilities of LLMs on the MoCA cognitive assessment.

- We designed a comprehensive, fine-grained evaluation framework that assesses model performance across the seven distinct cognitive subdomains of the MoCA, moving beyond a single aggregate score to provide a detailed capability profile.
- We conducted a systematic evaluation of various state-of-the-art multimodal LLMs, revealing their asymmetric capabilities and identifying key failure points through a novel, clinically-driven error taxonomy that offers insights for future model development.

RELATED WORK

Language Model Evaluations in Medicine

Existing datasets for evaluating LLMs in biomedicine [15] have primarily focused on verbal reasoning from static text. Benchmarks such as PubMedQA [8], MedQA [4], MedMCQA [5], and the medical portions of MMLU [10] have been instrumental in measuring the factual knowledge and descriptive reasoning capabilities of models. However, these datasets are mainly focused on qualitative reasoning from static documents, often in a multiple-choice question (MCQ) format. Additionally, the format of MCQs does not reflect the actual clinical settings where a single answer or response must be generated without any options provided, and more importantly, where the assessment is an interactive process rather than a one-off query. In this work, we introduce MoCA-Dialog, the first benchmark that measures the procedural and fine-grained reasoning capabilities of LLMs within a multimodal, dialogue-based clinical assessment, where the model must interpret and respond to a dynamic interaction.

Task-Oriented and Dialogue-Based Evaluation

Many efforts have been made to evaluate the capabilities of LLMs on more complex, procedure-based tasks. Datasets like GSM8k [11] and MATH [12] focus on multi-step mathematical problems. In the medical domain, MEDCALC-BENCH [14] represents a significant step forward by evaluating the ability of LLMs to perform quantitative calculations based on clinical notes, a task that more closely resembles a real-world clinical workflow. However, such datasets, while procedural, are not interactive and focus on quantitative computation. In contrast, MoCA-Dialog addresses the challenge of qualitative, interactive reasoning. While general dialogue-based benchmarks exist, they lack the domain-specific complexity, multimodal requirements, and fine-grained clinical scoring rules inherent in a neuropsychological assessment. As such, MoCA-Dialog is the first to combine a multimodal, dialogue-based format with the complex, nuanced reasoning required for a standardized clinical evaluation, providing a novel and challenging testbed for the next generation of clinical AI.

II. MOCA-DIALOG BENCHMARK

A. MoCA Task Analysis

The Montreal Cognitive Assessment (MoCA) consists of 11 core tasks, yielding a total score of 30 points across seven cognitive domains. Its complexity lies in the nuanced scoring criteria; for instance, in the Clock Drawing Test, the contour, numbers, and hands are each scored independently as one point. A detailed breakdown of all tasks and their official scoring rules is provided in Appendix A.

B. Multimodal Data Generation Pipeline

The construction of the MoCA-Dialog benchmark follows a systematic, three-stage generation pipeline, expanding from 50 expert-validated prototypical cases of cognitive impairment to a total of 5000 multimodal clinical assessment instances. All prototypical cases are derived from publicly available case reports on PubMed Central and anonymized patient vignettes provided by clinicians, ensuring that no identifiable personal health information is included. It is imperative to note that this dataset is designed for evaluating the capabilities of LLMs in medical cognitive assessment and is not intended for direct clinical diagnosis or medical decision-making.

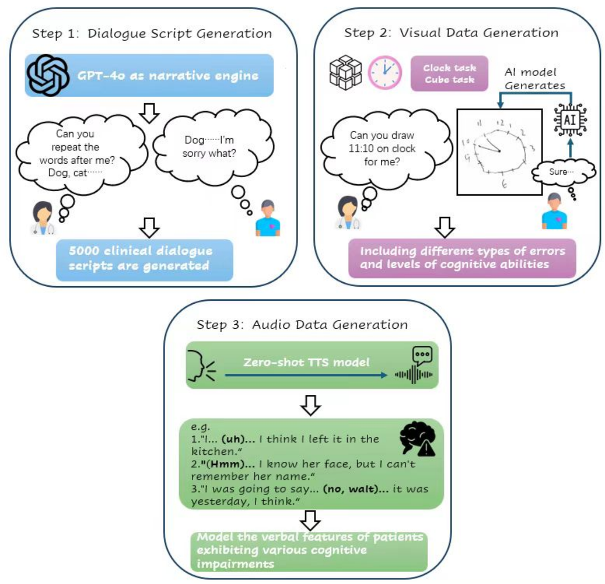

Fig. 1. The multimodal data generation pipeline for MoCA-Dialog, illustrating the parallel generation of dialogue scripts, visual assets, and audio responses from core clinical prototypes.

a) Dialogue Script Generation.: We employ GPT-4o as the core narrative engine to generate clinically plausible patient-clinician dialogues via a highly structured prompt framework. This framework defines three key dimensions for each case: cognitive impairment severity (mild, moderate, severe), etiological classifications (e.g., traumatic, cerebrovascular, psychiatric-related), and specific combinations of impaired cognitive domains. Each case is pre-configured with precise target error types, such as memory retrieval failures, executive function deficits, or impaired verbal fluency. To ensure authenticity, the model is instructed to strictly adhere to the standard MoCA administration protocol while naturally embedding characteristic linguistic phenomena—such as hesitations, self-corrections, and semantic substitutions—into the patient's responses. The generated scripts undergo automated consistency checks and sampled reviews by clinical experts to

TABLE I
COMPARISON OF MOCA-DIALOG WITH EXISTING LLM EVALUATION BENCHMARKS. COLUMNS DESCRIBE KEY CHARACTERISTICS: **TASK FORMAT** SPECIFIES THE PRIMARY INTERACTION STYLE; **EVALUATION FOCUS** INDICATES THE CORE CAPABILITY BEING TESTED; **INTERACTIVE** DENOTES IF THE TASK IS A DYNAMIC, MULTI-TURN PROCESS; **MULTIMODAL** INDICATES IF IT USES NON-TEXTUAL INPUTS.

Benchmark	Task Format	Evaluation Focus	Interactive	Multimodal
MedQA [4]	MCQ	Knowledge Recall	✗	✗
MedMCQA [5]	MCQ	Knowledge Recall	✗	✗
PubMedQA [8]	Static Q&A	Factual Reasoning	✗	✗
MMLU [10]	MCQ	General Knowledge	✗	✗
GSM8k [11]	Static Q&A	Math Reasoning	✗	✗
MATH [12]	Static Q&A	Math Reasoning	✗	✗
AgentMD [13]	Tool-Use	Clinical Tool Integration	✓	✗
MEDCALC-BENCH [14]	Procedure-based	Quantitative Reasoning	✗	✗
MoCA-Dialog (ours)	**Dialogue-based**	**Clinical Reasoning & Nuance**	✓	✓

ensure they are both compliant with assessment standards and accurately reflect the target cognitive profiles.

b) Visual Data Generation.: For visual tasks, we utilize the **Nano Banana** generation tool to directly create visual stimuli corresponding to pre-defined cognitive error types. For the Clock Drawing Test, we systematically generate image samples covering a spectrum of errors, including contour integrity defects (from mild distortion to complete unrecognizability), number placement errors (omissions, spatial misalignments, sequencing issues), and hand accuracy anomalies (missing hands, incorrect length differentiation, wrong time). For the Cube Copy task, we generate visual materials that include geometric distortions (line warping, angle deviations), spatial relationship errors (perspective anomalies, incorrect relative positioning), and deficits in structural integrity. Through fine-grained prompt engineering, we achieve graded control over the severity of errors, ensuring a continuous range of samples from mild to severe for each error type, thereby providing rich visual cues for model evaluation.

c) Audio Data Generation.: We use the zero-shot TTS model **CosyVoice2** to simulate the speech characteristics of patients with cognitive impairments through multi-level parametric control. At the level of basic speech parameters, we configure different speech rates (70%-90% of normal), prosody patterns (flattened intonation, irregular pauses), and articulation clarity based on the impairment type and severity. At the paralinguistic level, we systematically inject three classes of typical phenomena: filler words (e.g., "um," "uh" as markers of hesitation), speech disfluencies (word repetitions, phrase breaks), and self-correction behaviors. Crucially, we configure specific phonetic patterns for different cognitive impairments, such as prolonged pauses for patients with memory deficits or interruptions in speech flow for those with executive dysfunction, thus acoustically reconstructing the complexity of the clinical encounter.

III. DATASET CHARACTERISTICS

Table II shows the statistics of the MoCA-Dialog benchmark and its different clinical prototype sub-types. The dataset is structured around 50 core prototypes, which are grouped into broader etiological and phenomenological categories. Each instance is designed with a varying number of target cognitive errors (from 1 to 8) to ensure a wide range of difficulty.

Our dataset evaluates three distinct capabilities required for clinical cognitive assessment:

a) (1) Recall and application of clinical protocol knowledge.: The first required capability is to successfully recall and apply the complex, often nuanced scoring rules of the MoCA. As shown in Table II, our benchmark covers a wide range of clinical prototypes, each requiring the model to apply the rules to different error patterns. Hence, LLMs are challenged to know every detail about the MoCA protocol to correctly score a dynamic conversation.

b) (2) Extraction of relevant multimodal and linguistic cues.: The second required capability is the extraction of correct diagnostic information from a multi-turn dialogue, given the noise of natural conversation and the subtlety of cognitive impairment markers. LLMs are required to interpret not just the literal text, but also paralinguistic features (hesitations, self-corrections simulated in the text and audio) and visual-spatial information (from images and their verbal descriptions). The clinical context complicates such extractions, requiring the model to differentiate between a knowledge gap (e.g., "I don't know what a rhino is") and a retrieval deficit (e.g., "I know what it is... that big animal with a horn... I can't find the word").

c) (3) Synthesis and clinical reasoning for interpretation.: The third required capability is the synthesis of information from multiple tasks to form a coherent clinical interpretation. While some tasks test isolated abilities, a final clinical judgment requires multi-step reasoning to identify a pattern of impairment (e.g., recognizing that poor delayed recall combined with impaired clock drawing is highly suggestive of a specific dementia profile). LLMs must derive a clinically meaningful cognitive profile, moving beyond individual item scores.

A. Settings

To establish the baseline performance on MoCA-Dialog, we experiment with four state-of-the-art multimodal large language models. We categorize them into two groups: **Proprietary Models** include Gemini 2.5 Pro, GPT-4o, and Claude

TABLE II
STATISTICS OF THE MOCA-DIALOG BENCHMARK. INST.: INSTANCE; AVG.: AVERAGE.

Cognitive Profile Prototype	# Proto.	# Inst.	Avg. Dialogue Turns	Avg. Target Errors	Example Simulated Cue
Memory-centric Impairments					
Amnestic Type (e.g., AD)	15	2000	29.5	4.2	”I can’t recall the words... was one a flower?”
Executive/Vascular Impairments					
Executive Dysfunction Type (e.g., FTD)	10	1000	31.2	5.1	”1-A-2-B... then... wait... C? No...” (TMT self-correction)
Vascular Type (Mixed deficits)	10	1000	30.1	4.5	(Simulated slowed speech rate and word-finding pauses)
Other Impairments					
Psychiatric-related (e.g., Depression)	5	500	28.1	2.8	”I don’t know, my mind is blank.” (Low effort/affect)
Traumatic Brain Injury (TBI) Type	5	500	32.5	5.5	(Attention deficits, poor verbal fluency)
Overall	**50**	**5000**	**30.2**	**4.6**	–

3.5; and **Open-Source Models**, represented by LLaVA-NeXT-34B.

Similarly, we consider four distinct prompting strategies of increasing complexity to systematically evaluate model capabilities:

- **Zero-shot** The model is prompted with the multimodal inputs (dialogue, images, audio transcription) and asked to directly output the scores without any examples or explicit rules.
- **Zero-shot + Instructions** In this setting, the model is provided with the complete, official MoCA administration and scoring guide within the prompt, in addition to the case inputs. This tests the model’s ability to follow complex instructions.
- **One-shot** In this setting, the model is provided with a single, complete exemplar instance (including inputs, correct scores, and reasoning), but not the general scoring guide. This tests the model’s in-context learning from a specific example.
- **One-shot + Instructions** This is the most comprehensive setting, where the LLM is provided with both the scoring guide and a complete exemplar.

Based on the nature of our benchmark, we use two main evaluation metrics: (1) For overall performance, we calculate the **Total Score Mean Absolute Error (MAE)**, where a lower value is better (range 0-30). (2) To assess fine-grained capabilities, we compute the **Macro F1-Score** for three distinct cognitive domain groups: **Visuospatial/Executive**, **Attention/Memory**, and **Language/Abstraction**. A higher F1-Score is better.

B. Main Results

Table III presents our evaluation results of the four multimodal LLMs on the MoCA-Dialog test set across the four prompting strategies. From the table, we can observe a clear and consistent trend: model performance progressively improves with more sophisticated prompting. Among all LLMs compared, Gemini 2.5 Pro achieves the best performance in all settings, particularly in the most comprehensive **One-shot + Instructions** setting, where it reaches a Total Score MAE of 2.85. Similar patterns of improvement can be observed in all other models, though their absolute performance varies.

In addition to the general trend, the table also shows how different types of LLMs perform on our benchmark. While Gemini 2.5 Pro performs the best in our evaluation, GPT-4o shows competitive performance that is close, especially in language-related domains. Claude 3.5 follows, demonstrating solid capabilities but lagging in the most complex reasoning tasks. There is a significant performance gap between the proprietary models and the open-source LLaVA-NeXT, highlighting the current challenges for open-source multimodal models in this highly specialized domain.

It can also be observed that the results for different prompting strategies present distinct patterns. The most significant performance leap for all models occurs when moving from **Zero-shot** to **Zero-shot + Instructions**. This suggests that a primary failure mode for LLMs is the lack of specific, in-context knowledge of the complex MoCA scoring protocol, which can be substantially mitigated by providing the official guide. The addition of a one-shot example further refines performance, especially on nuanced tasks. For instance, the Visuospatial/Executive F1-score for Gemini 2.5 Pro sees a notable 12% relative improvement between the ‘Zero-shot + Instr.‘ and ‘One-shot + Instr.‘ settings. This reflects that the exemplar helps the model understand how to apply the rules to complex, ambiguous inputs (e.g., interpreting a patient’s description of a distorted clock), a skill that instructions alone cannot fully convey. Such an analysis enables us to have insights into the capabilities and limitations of LLMs on multimodal clinical assessment tasks.

IV. DISCUSSION

In this section, we provide an in-depth analysis of the errors made by LLMs on MoCA-Dialog.

TABLE III
MoCA-Dialog accuracy of different multimodal systems. All F1-Scores are percentages. MAE is the Mean Absolute Error for the total score (lower is better). F1 scores for cognitive domains are Macro F1-Scores (higher is better). Best performance in each category is highlighted in **bold**.

Model	Size	Total Score MAE (↓)	Visuospatial/Exec F1 (↑)	Attention/Memory F1 (↑)	Language/Abstract F1 (↑)
Zero-shot Prompting					
Gemini 2.5 Pro	N/A	8.11	58.31	71.45	49.02
GPT-4o	N/A	9.25	55.19	69.82	51.15
Claude 3.5	N/A	11.04	49.53	65.11	44.38
LLaVA-NeXT	34B	16.58	31.08	45.94	28.17
Zero-shot + Instructions Prompting					
Gemini 2.5 Pro	N/A	4.38	75.88	88.23	69.46
GPT-4o	N/A	5.10	72.45	87.91	70.53
Claude 3.5	N/A	6.92	66.71	82.50	61.22
LLaVA-NeXT	34B	10.83	49.62	61.37	42.85
One-shot Prompting					
Gemini 2.5 Pro	N/A	5.57	71.02	84.18	65.33
GPT-4o	N/A	6.31	68.99	83.55	67.24
Claude 3.5	N/A	8.24	62.15	78.06	58.91
LLaVA-NeXT	34B	12.50	45.14	58.22	39.05
One-shot + Instructions Prompting					
Gemini 2.5 Pro	N/A	2.85	85.12	94.31	79.58
GPT-4o	N/A	3.49	82.05	93.88	78.64
Claude 3.5	N/A	4.98	76.83	90.17	70.19
LLaVA-NeXT	34B	8.95	58.44	70.56	51.72

A. What types of errors can LLMs make on MoCA-Dialog?

We categorize four types of errors that LLMs can make in our dataset: **Type A (Knowledge Errors)** the model does not have the correct knowledge of the MoCA scoring rules (e.g., forgetting the educational adjustment); **Type B (Cue Extraction & Interpretation Errors)** the model incorrectly extracts or misinterprets key verbal or multimodal cues from the patient's response; **Type C (Reasoning Errors)** the model correctly scores individual items but fails to synthesize them into a coherent clinical profile or diagnosis; and **Type D (Other Errors)** all other cases of errors. Specific examples of the first three error types are shown in Table IV.

It should be noted that these errors are not independent of each other. For example, LLMs usually recall the relevant scoring rule first (Type A), then interpret the patient's response (Type B), and finally conduct the clinical reasoning (Type C). If the model recalls the rule incorrectly, it is highly likely that it will misinterpret the subsequent cues. Hence, we only consider the earliest error if there are multiple error types (e.g., if the model has error types A and B, then the error type will be A).

B. What errors do different LLMs make?

To analyze the errors made by different LLMs, we utilize GPT-4o to classify their error types by comparing the LLM output to the ground truth in MoCA-Dialog. We manually evaluate the annotations of 200 randomly sampled explanation errors and find the accuracy of the GPT-4o error classifier to be 91%. As such, we apply it to analyze the mistakes of all prompting results.

Table V shows the distribution of error types in different settings. Under the zero-shot setting, most of the errors (more than 60% for all models) belong to Type A, suggesting that recalling the correct scoring rules for the MoCA is the biggest challenge when no exemplar is provided. While Type A error is dominant under the zero-shot setting, its error rate varies in different LLMs, e.g. 0.68 in LLaVA-NeXT and 0.39 in Gemini 2.5 Pro. Such a difference reflects the diverse levels of clinical knowledge acquired by various LLMs.

Unlike the distributions in the zero-shot setting, errors that occurred with the one-shot prompting have less than 40% being categorized as Type A, which is consistently observed in different LLMs. This shows the effectiveness of the one-shot exemplar in providing the background rules needed for the assessment. With the decrease in Type A errors, more Type B and Type C errors are captured in the wrong answers. This reveals the deficiencies of current LLMs in cue interpretation and clinical reasoning, which are required capabilities to perform real-world clinical assessments.

C. Qualitative Case Study: Failure in Multimodal Visuospatial Reasoning

A consistent area of weakness across all tested models is the interpretation of visuospatial tasks, particularly the Clock Drawing Test. This task requires the model to integrate a visual input (the drawing) with a specific domain rule set (the MoCA scoring criteria for a clock). Many models exhibit a profound failure in applying a numerical-to-semantic mapping specific to this domain.

TABLE IV
EXAMPLES OF THREE ERROR TYPES IN MOCA-DIALOG. THE INCORRECT RATIONALE AND EVIDENCE ARE HIGHLIGHTED.

Type	Question	Patient Dialogue Snippet	Error of a Zero-shot LLM
A - Knowledge Error	What is the patient's final MoCA score?	*(Patient Profile: 11 years of education)* Patient scores 25 points across all tasks in the dialogue.	The model correctly sums the raw score to 25. The final score is reported as 25. # The model forgot the rule to add 1 point if education is less than 12 years. The correct score should be 26.
B - Cue Extraction & Interpretation Error	What is the score for Abstraction?	*Doctor:* "Tell me how a watch and a ruler are alike." *Patient:* "Well, they both have numbers on them."	The model assigns 1 point for abstraction. Rationale: "The patient provided a valid similarity between the two items." # This is a concrete similarity, not abstract. Correct answer is 'measuring instruments', score should be 0.
C - Reasoning Error	Generate a cognitive profile summary.	*(Patient scores 0/5 on Delayed Recall task, but scores well on Attention and Language tasks.)*	The model's generated profile: "The patient shows good attention and language abilities. Cognitive functions appear largely intact." # The model completely fails to mention the severe memory impairment, a critical diagnostic finding, indicating a reasoning failure to connect the score to its clinical meaning.

TABLE V
ERROR TYPE DISTRIBUTION OF LLMS ON MOCA-DIALOG. NUMBERS IN PARENTHESES DENOTE THE RELATIVE PROPORTIONS. ARROWS INDICATE THE PROPORTION CHANGES FROM ZERO-SHOT TO ONE-SHOT LEARNING.

Model	Type A Error	Type B Error	Type C Error	Type D Error	Error Rate
Zero-shot Prompting					
LLaVA-NeXT	0.68 (79%)	0.11 (13%)	0.05 (6%)	0.02 (2%)	0.86
Claude 3.5	0.51 (68%)	0.15 (20%)	0.07 (9%)	0.02 (3%)	0.75
GPT-4o	0.43 (61%)	0.18 (26%)	0.08 (11%)	0.01 (2%)	0.70
Gemini 2.5 Pro	0.39 (57%)	0.20 (29%)	0.09 (13%)	0.01 (1%)	0.68
One-shot + Instructions Prompting					
LLaVA-NeXT	0.25 (38%↓)	0.26 (40%↑)	0.13 (20%↑)	0.02 (2%−)	0.66
Claude 3.5	0.17 (31%↓)	0.23 (42%↑)	0.13 (24%↑)	0.02 (3%−)	0.55
GPT-4o	0.12 (26%↓)	0.22 (48%↑)	0.10 (22%↑)	0.01 (2%−)	0.46
Gemini 2.5 Pro	0.09 (22%↓)	0.21 (51%↑)	0.11 (27%↑)	0.00 (0%↓)	0.41

TABLE VI
CASE STUDY OF GEMINI 2.5 PRO'S ERROR ON THE CLOCK DRAWING TEST.

Criterion	Ground Truth Analysis (Score: 2/3)	Gemini 2.5 Pro Rationale (Score: 2/3)
Contour	Correct (1/1): The clock face is a closed circle, minor imperfections are acceptable.	Correct (1/1): "The clock face... is a roughly closed circle... this item receives 1 point."
Numbers	Correct (1/1): All numbers are present, in the correct order, and in the appropriate quadrants.	Correct (1/1): "All numbers from 1 to 12 are present and reasonably placed... this item receives 1 point."
Hands	**Incorrect (0/1)**: Although two hands are present, the minute hand points to the number '10', representing 50 minutes, not 10 minutes as required. This is a critical error.	**Incorrect (0/1)**: "The scoring criteria for the hands have not been met. The minute hand correctly points to the number 2 (representing 10 minutes). The hour hand is pointing incorrectly to 11..."

Consider the following case where the patient's drawing shows the minute hand pointing directly at the number '10'. The correct interpretation is 50 minutes. We provided the image and the following prompt to Gemini 2.5 Pro: 'Score the Clock Drawing Test based on the provided image for the time 11:10.'

As shown in Table VI, Gemini 2.5 Pro correctly evaluates the contour and numbers. However, it makes a critical reasoning error when evaluating the hands. The model's rationale states that the minute hand points to '2', even though the image clearly shows it pointing to '10'. This is not a failure of visual perception (the model knows what the number '10' looks like), but a failure of domain-specific reasoning. The model appears to be "hallucinating" the correct answer ('2' for 10 minutes) because it is reasoning backwards from the requested time ('11:10') rather than interpreting the visual evidence

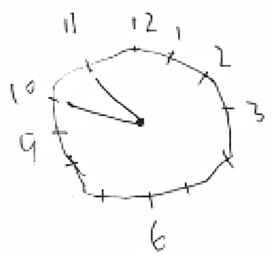

Fig. 2. Case study for clock drawing task for the time 11:10

presented. This reveals a significant limitation: while LLMs possess general world knowledge, they struggle to inhibit that knowledge and strictly apply the specialized, procedural rules of a clinical test, leading to critical errors in evaluation.

V. Conclusion

In conclusion, this study introduces MoCA-Dialog, the first large-scale, multimodal benchmark designed to evaluate the capabilities of LLMs for the complex, interactive task of clinical cognitive assessment. Our evaluations show that while state-of-the-art LLMs like Gemini 1.5 Pro exhibit potential, none are reliable enough for autonomous clinical use. The error analysis highlights critical areas for improvement, revealing that the primary failure modes stem not from a lack of clinical knowledge but from profound deficits in applying domain-specific reasoning and interpreting nuanced multimodal and linguistic cues. We hope our work serves as a call to action to further improve the procedural and fine-grained clinical reasoning capabilities of LLMs, making them safer and more suitable for the complex, interactive workflows of modern medicine.

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